



UTM

UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
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BACHELOR OF COMPUTER SCIENCE (BIOINFORMATICS)

SECV 2113 – HUMAN COMPUTER INTERACTION

EV CHARGING STATIONS TRACKER

TASK 2

GROUP: NOBODY

GROUP MEMBERS	MATRIC NO
TAN ZHAO HONG	A23CS0188
CHIN PEI WEN	A23CS0065
KOO XUAN	A23CS0300
LING YU QIAN	A23CS0301
NAZATUL NADHIRAH BINTI SABTU	A23CS0144

Lecturer's name : Dr. Nur Zuraifah Syazrah binti Othman

Title: Gathering Requirements - Task Analysis

Introduction

Electric vehicle drivers are facing challenges to maintain their vehicles' charge, especially on long journeys or in areas far from the city which have limited charging stations. To solve this problem, EV charging stations tracker applications become useful tools for various stakeholders, including EV car owners, fleet managers, delivery drivers and travelers and road trippers.

BUT U EV Charging Stations Tracker App is tested to understand how it can improve upon existing systems like BYD and ParkEasy. The observations focus on 3 critical tasks which is choosing or finding a charging station, monitoring the charging process and offering multiple payment options. Firstly, users begin with choosing or finding a charging station which involves location nearby stations based on compatibility and availability. Then, monitoring the charging process to allow users to keep track for the status and progress of their vehicle's charge in real time. Finally, the application offers multiple payment options to make users flexible and convenient during the payment process. These tasks ensure a smooth EV charging experience for all stakeholders, catering to their diverse needs and preferences.

Derivation of HTA

HTA for Task 1 - Choosing / Find Charging Station

Type of User 1 : EV Car Owner

{Link/Embedded Video on User 1 doing Task 1 (e.g., research student testing tasks 1)}

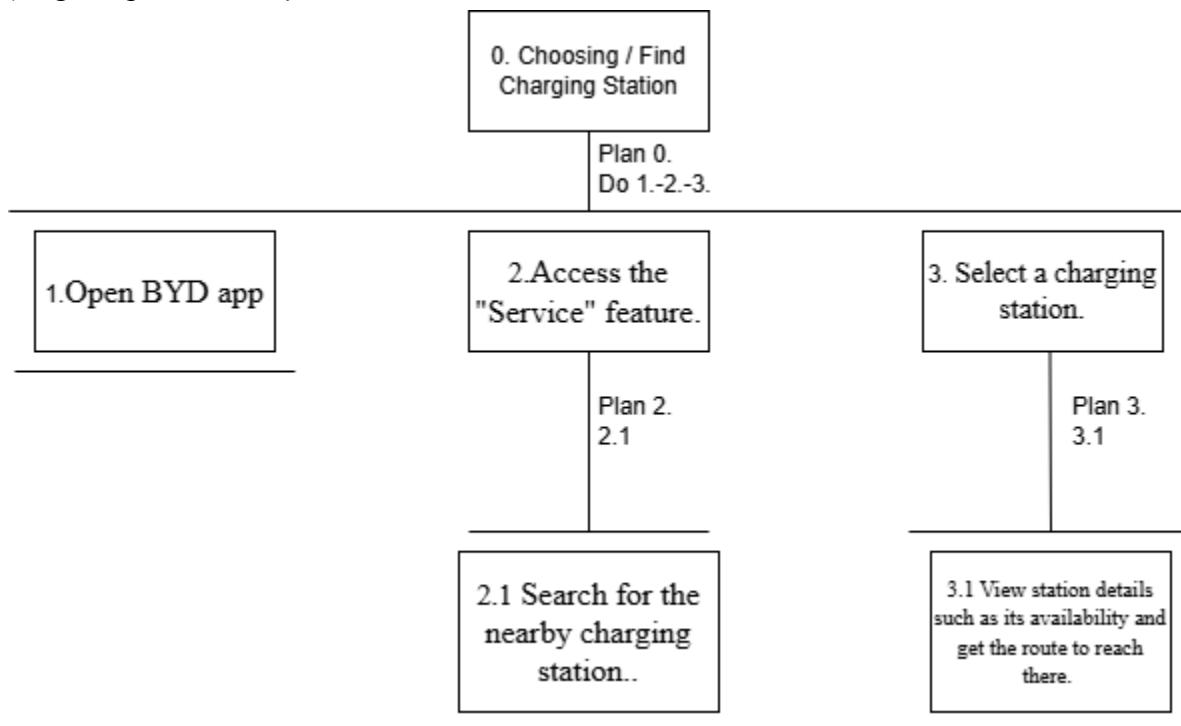
<https://youtu.be/3sLMp0N1Mt4>

HTA for User 1

(textual presentation)

1. Open BYD app.
2. Access the "Service" feature.
 - 2.1 Search for the nearby charging station.
3. Select a charging station.
 - 3.1 View station details such as its availability and get the route to reach there.

(diagram presentation)



Type of User 2 : EV Car Company Manager

{Link/Embedded Video on User 2 doing Task 1(e.g., doctor testing task 1)}

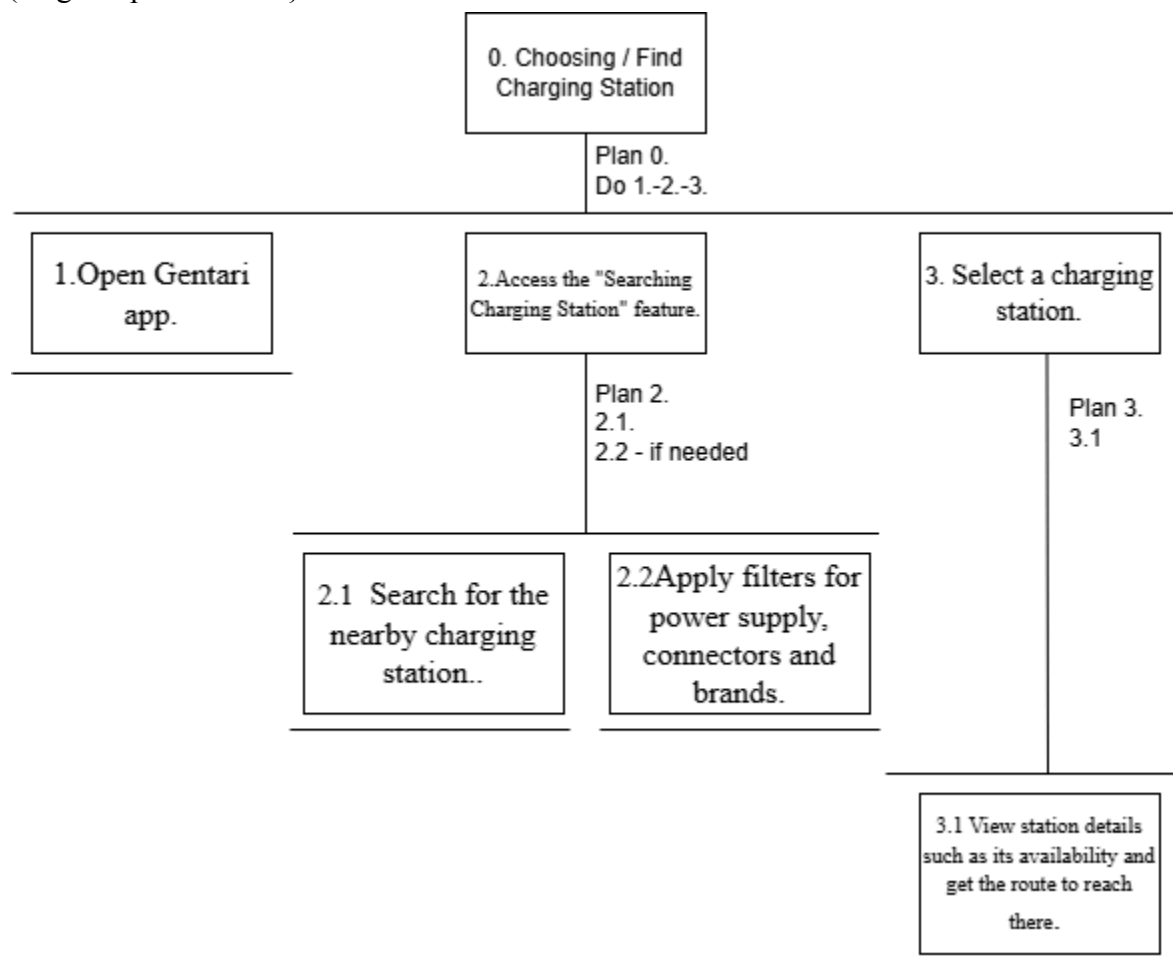
<https://youtu.be/W2RqlbAlrOs>

HTA for User 2

textual presentation

1. Open Gentari app.
2. Access the "Searching Charging Station" feature.
 - 2.1 Apply filters for power supply, connectors and brands.
 - 2.2 Search for the nearby charging station.
3. Select a charging station.
 - 3.1 View station details such as its availability and get the route to reach there.

(diagram presentation)



Type of User 3 : Traveler / Road Tripper

{Link/Embedded Video on User 3 doing Task 1(e.g., Housewife testing tasks 1)}

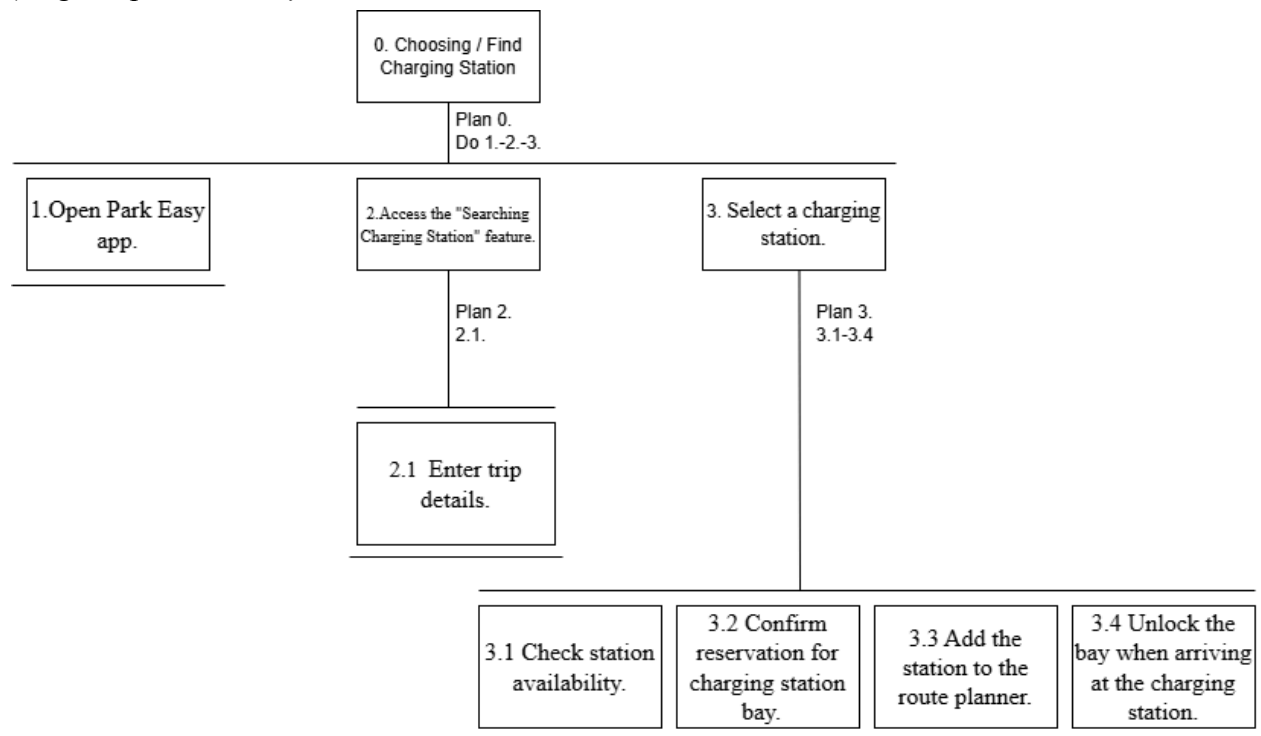
https://youtu.be/wEMSOK_oLqI

HTA for User 3

(textual presentation)

1. Open Park Easy app.
2. Access the "Search Charging Station" feature.
 - 2.1 Enter trip details.
3. Select a charging station.
 - 3.1 Check station availability.
 - 3.2 Confirm reservation for charging station bay.
 - 3.3 Add the station to the route planner.
 - 3.4 Unlock the bay when arriving at the charging station.

(diagram presentation)



Findings from the HTAs for Task 1

EV car owners, company and travelers encountered no significant issues when searching for nearby charging stations using the apps. This app has a map-based interface to make it easy for users to locate stations and filter by charger type and availability. However, travelers sometimes faced difficulties when searching for stations with nearby amenities such as rest areas and shopping malls. To address this problem, we suggest introducing additional filter options for nearby amenities and integrating third-party information about facilities available at each charging location.

HTA for Task 2 - Offer multiple payments option

Type of User 1 : EV Car Owner

{Link/Embedded Video on User 1 doing Task 1 (e.g., research student testing tasks 1)}

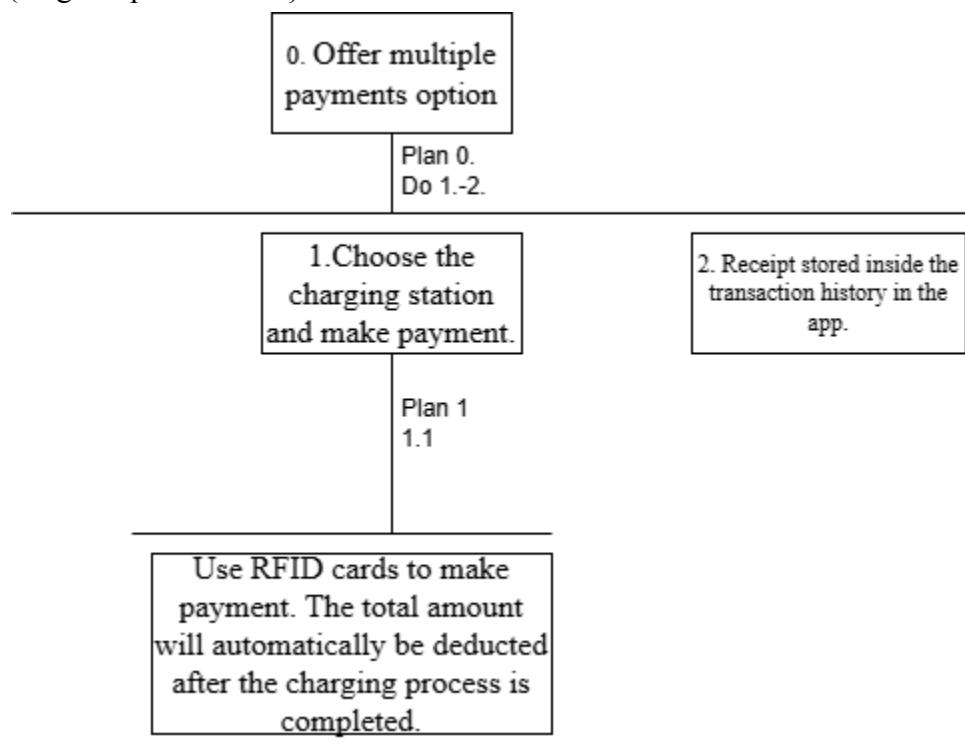
<https://youtu.be/HH7fluecrpo>

HTA for User 1

(textual presentation)

1. Choose the charging station and make payment.
 - 1.1 Use RFID cards to make payment. The total amount will automatically be deducted after the charging process is completed.
2. Receipt stored inside the transaction history in the app.

(diagram presentation)



Type of User 2 : EV Car Company Manager

{Link/Embedded Video on User 2 doing Task 1(e.g., doctor testing task 1)}

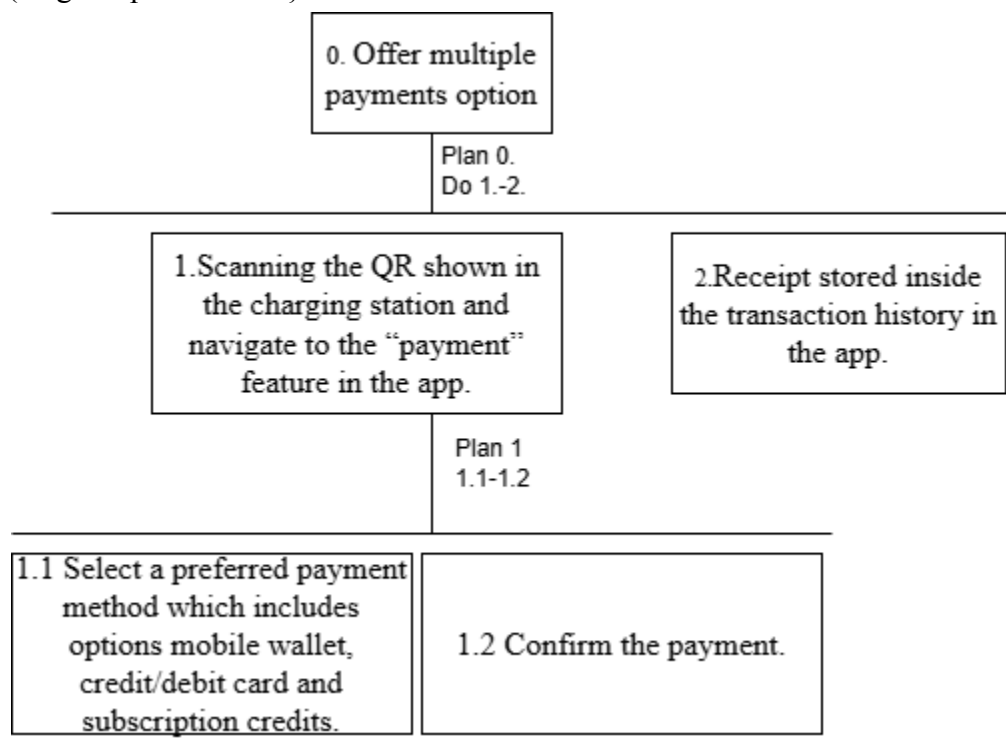
<https://youtu.be/S8gZrRz-VKk>

HTA for User 2

(textual presentation)

1. Scanning the QR shown in the charging station and navigate to the “payment” feature in the app.
 - 1.1 Select a preferred payment method which includes options mobile wallet, credit/debit card and subscription credits.
 - 1.2 Confirm the payment.
2. Receipt stored inside the transaction history in the app.

(diagram presentation)



Type of User 3 : Traveler / Road Tripper

{Link/Embedded Video on User 3 doing Task 1(e.g., Housewife testing tasks 1)}

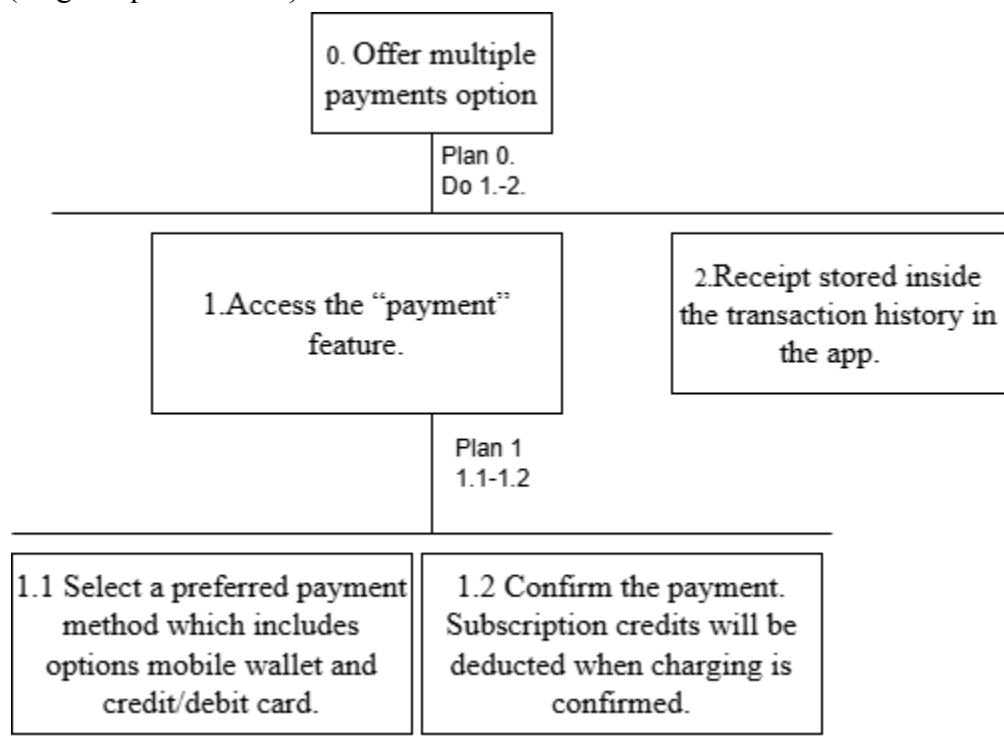
<https://youtu.be/vS4XwPAYstA>

HTA for User 3

(textual presentation)

1. Access the “payment” feature in the Park Easy app.
 - 1.1 Select a preferred payment method which includes options mobile wallet and credit/debit card.
 - 1.2 Confirm the payment. Subscription credits will be deducted when charging is confirmed.
2. Receipt stored inside the transaction history in the app.

(diagram presentation)



Findings from the HTAs for Task 2

All user types, including EV car owners, company, and travelers/road trippers’ HTA are the same, they encountered no significant issues while completing the payment process. The steps to choose a charging station, select a payment method, and confirm the transaction were smooth and straightforward for all users.

HTA for Task 3 - Track charging progress

Type of User 1 : EV Car Owner

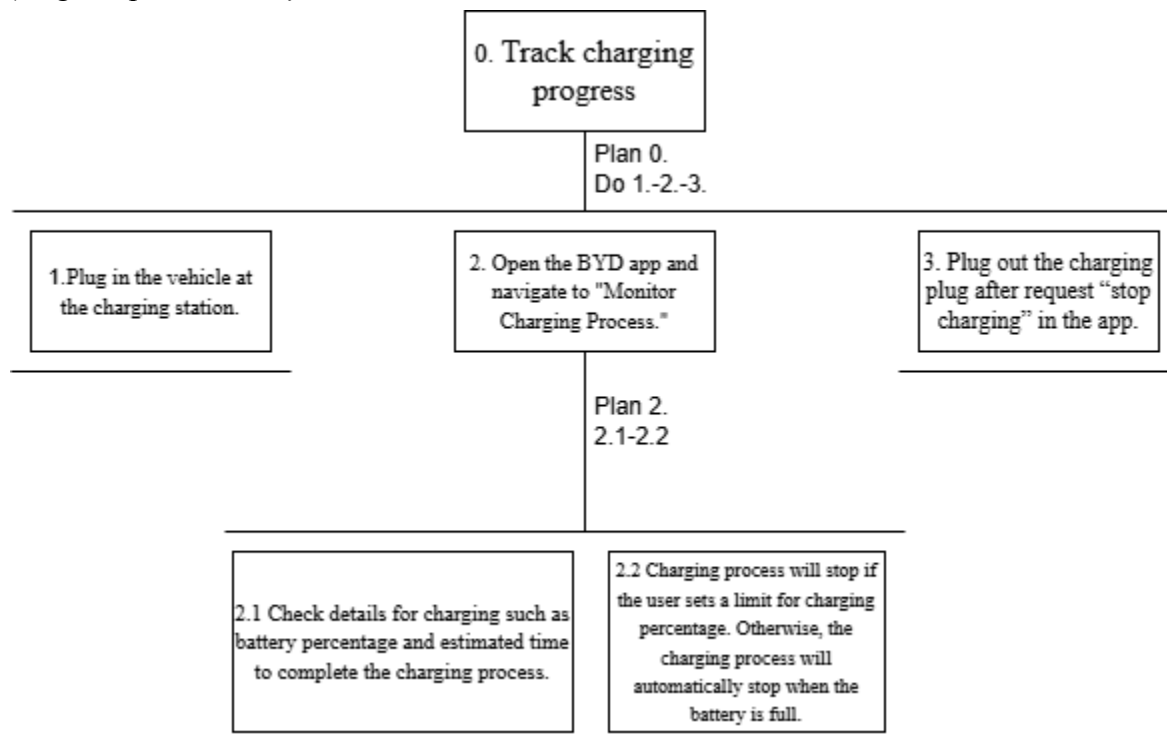
{Link/Embedded Video on User 1 doing Task 1 (e.g., research student testing tasks 1)}

<https://youtu.be/gvVKC6TcbBM>

HTA for User 1

1. Plug in the vehicle at the charging station.
2. Open the BYD app and navigate to "Monitor Charging Process."
 - 2.1 Check details for charging such as battery percentage and estimated time to complete the charging process.
 - 2.2 Charging process will stop if the user sets a limit for charging percentage. Otherwise, the charging process will automatically stop when the battery is full.
3. Plug out the charging plug after request "stop charging" in the app.

(diagram presentation)



Type of User 2 : EV Car Company Manager

{Link/Embedded Video on User 2 doing Task 1(e.g., doctor testing task 1)}

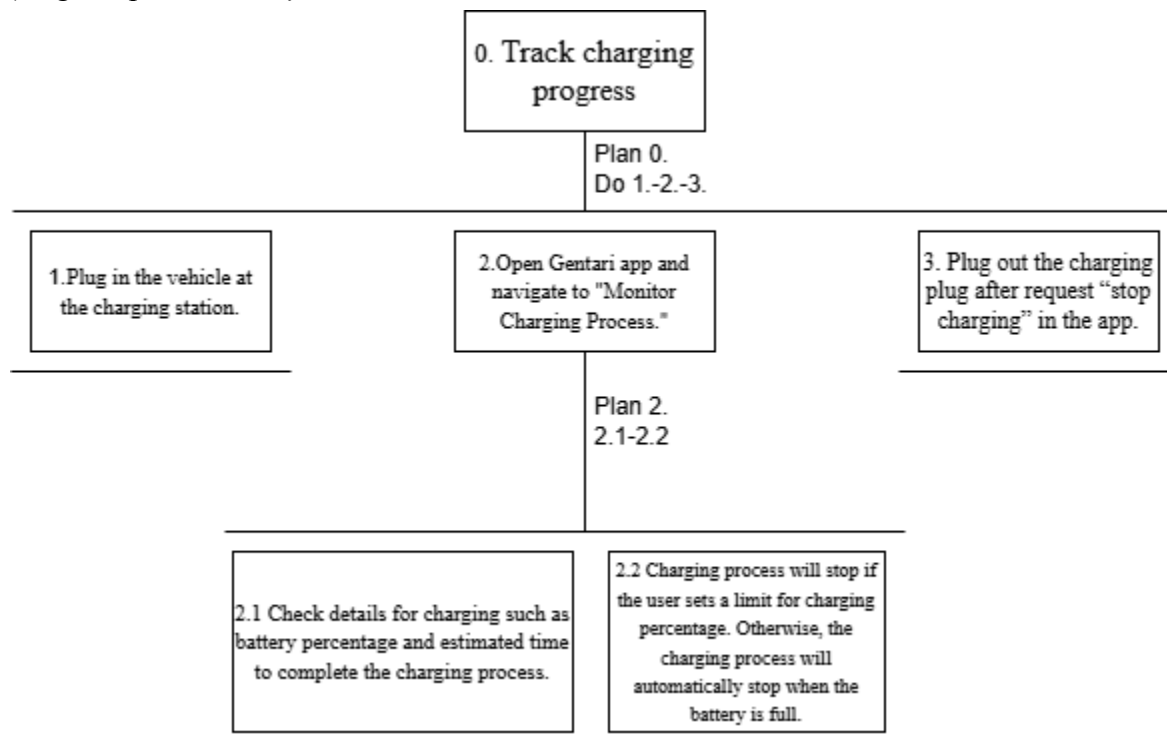
<https://youtu.be/e3cvuHBH-8s>

HTA for User 2

(textual presentation)

1. Plug in the vehicle at the charging station.
2. Open the BYD app and navigate to "Monitor Charging Process."
 - 2.1 Check details for charging such as battery percentage and estimated time to complete the charging process.
 - 2.2 Charging process will stop if the user sets a limit for charging percentage. Otherwise, the charging process will automatically stop when the battery is full.
3. Plug out the charging plug after request "stop charging" in the app.

(diagram presentation)



Type of User 3 : Traveler / Road Tripper

{Link/Embedded Video on User 3 doing Task 1(e.g., Housewife testing tasks 1)}

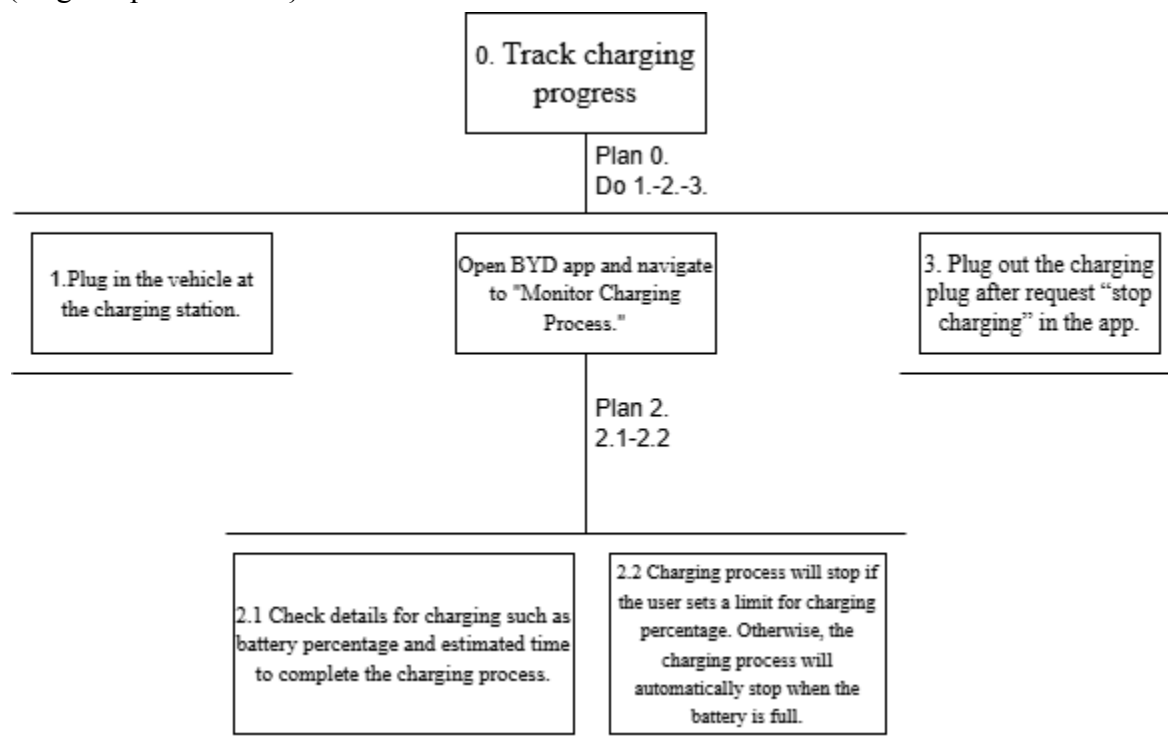
<https://youtu.be/Ik2sXpxEpPI>

HTA for User 3

(textual presentation)

1. Plug in the vehicle at the charging station.
2. Open the BYD app and navigate to "Monitor Charging Process."
 - 2.1 Check details for charging such as battery percentage and estimated time to complete the charging process.
 - 2.2 Charging process will stop if the user sets a limit for charging percentage. Otherwise, the charging process will automatically stop when the battery is full.
3. Plug out the charging plug after request "stop charging" in the app.

(diagram presentation)



Findings from the HTAs for Task 3

In this task, the HTA of EV Car Owners, companies and travelers are the same and they did not face major problems while monitoring the charging process. However, some users may report delays in real time updates when charging session stopped or when there is out of power for the stations. These may cause confusion for those users. Thus, the app could implement instant notifications whenever a charging session is interrupted or completed.

Design requirements

After observing the use of the existing system by various users, we have identified several design requirements for the But U system :

- Add filters for nearby amenities to enhance the user's ability to choose stations based on their preferences.
- Provide recommendations for nearby attractions or activities during the charging process especially for travelers.
- Implement instant alerts/ notifications for users when a charging session is interrupted, completed or there are power issues at the station.
- Ensure the system supports diverse payment options for flexibility and convenience.
- Including multilingual support based on our country is a multi-ethnic country.
- Introduce incentives such as challenges to win rewards or discounts for consistent app usage or eco-friendly driving habits.

By observing the different feedback and difficulties encountered by each user during these tasks, we have identified areas to improve the existing system. These design requirements aim to resolve the identified issues and help us create a better system that meets the needs of users in terms of User Interface(UI) and User Experience(UX) design.

User Analysis

Title: Gathering Requirements - User Analysis

Proposed Tasks

- **Choosing/Finding a Charging Station**
Users can search and select a charging station based on compatibility, distance, and availability. The app allows users to filter by charger type, speed, and nearby amenities, ensuring users can find the most suitable charging station for their needs. The map-based interface helps users locate charging stations quickly and plan routes efficiently.
- **Make Payment**
Users use their preferred payment method, including mobile wallets, credit/debit cards, and subscriptions credits to complete the payment process for charging sessions. The app will also securely process and automatically store the receipts in the transaction history for future reference.
- **Charge the vehicle and monitor charging progress**
Users plug in the vehicle and start charging. Users monitor the charging process of their vehicle in real-time through the app. The app provides details like the current battery percentage, estimated time to complete charging and any alerts regarding the charging session. Users can set their limitations of battery level and the charging process will stop when the battery reaches the limit, otherwise the system will default to charging the battery fully. After the user knows the time estimation for their desired battery level, then the user can choose either want to have a fast charging or normal charging based on the given estimated time.

Persona: Olivia Tan - The Urban Commuter

Olivia Tan, a 28-year-old marketing executive, resides in the bustling city of Kuala Lumpur. She is an environmentally conscious individual who recently transitioned to driving an electric vehicle (EV) as part of her commitment to sustainability. Olivia's weekdays are packed with meetings and errands, requiring frequent city commutes. Her busy schedule leaves little room for unexpected disruptions, so she values reliability and efficiency in her daily routines. She also appreciates intuitive technology that can simplify her life, especially mobile applications that provide practical solutions to her transportation needs.

Despite the convenience of EVs, Olivia often finds herself stressed when searching for available charging stations during peak hours. She struggles to locate nearby chargers with compatible ports or sufficient speed for a quick top-up. Payment inconsistencies at different charging

stations further frustrate her. Olivia seeks a seamless solution that helps her efficiently find and reserve charging stations, offers real-time updates on availability, and simplifies the payment process—all while fitting into her hectic lifestyle.

2. Persona: Daniel Lee - The EV Car Company Manager

Daniel Lee is a 38-year-old sales manager at a leading electric vehicle (EV) dealership in Kuala Lumpur. He has over 10 years of experience in the automotive industry. Three years ago, he shifted to managing EV sales, aligning with the company's focus on sustainable transportation. Daniel leads a team of sales representatives, plans marketing strategies, and ensures customers enjoy a seamless buying process. He is passionate about the rising interest in EVs and takes pride in educating customers on their benefits, such as cost savings, environmental advantages, and innovative features.

Despite the growing demand, Daniel faces challenges in addressing customer concerns. Many customers worry about the charging infrastructure when transitioning from petrol-powered cars. They frequently ask about station availability, locations, and compatibility with different EV models. Daniel needs tools like the "BUT U" app to provide accurate, real-time information on charging networks. This tool would help him resolve issues like range anxiety and inconsistent payment options. He also sees the app as a way to enhance customer satisfaction, building trust and confidence in EV technology and his dealership.

3. Persona: Siti Nuraini - The Road Tripper

Siti Nuraini is a 28-year-old travel enthusiast from Johor Bahru who enjoys exploring Malaysia's scenic routes in her electric vehicle. As someone who loves the freedom of road trips, she frequently embarks on weekend getaways with friends or solo journeys to new destinations. Siti values convenience, comfort, and experiences that allow her to stay connected with nature while reducing her carbon footprint. She uses travel apps to plan her routes and find hidden gems, such as scenic spots or local eateries, along the way.

While Siti enjoys the perks of EV travel, such as cost savings and environmental benefits, she often experiences range anxiety when driving in unfamiliar areas. She struggles to find charging stations with amenities like rest areas or restaurants, which are essential for her long trips. Siti looks for an app that not only helps her locate reliable charging stations but also enhances her journey by suggesting nearby attractions or activities, making her road trips both stress-free and memorable.

4. Scenario

Scenario 1: Finding a Charging Station (Olivia Tan - The Urban Commuter)

On a busy Monday morning, Olivia is preparing for an important client meeting across the city. As she heads out, she notices that her EV's battery is down to 20%, which might not last her entire trip. By using our "BUT U" app on her phone, she can select the "Find Charging Station" feature, and apply filters for fast chargers along her route. Within seconds, the app will suggest a nearby station and confirm that a slot is available. Olivia can reserve the spot and check additional details, such as the estimated charging time and nearby amenities.

Upon arriving at the station, Olivia finds her reserved slot waiting for her which thanks to the app's real-time updates. She connects her car to the charger and pays seamlessly using the app's integrated payment system. While her car charges, Olivia can review her meeting notes and will receive a notification when her EV's battery reaches the desired level. This streamlined process allows her to stay on schedule, reinforcing her confidence in driving an EV for her fast-paced lifestyle.

Scenario 2: Supporting Customers with EV Adoption (Daniel Lee – The EV Sales Manager)

Daniel meets Sarah, a potential customer that is interested in buying her first electric vehicle (EV). However, Sarah is hesitant due to concerns about charging infrastructure. She frequently takes long-distance trips and worries about finding reliable charging stations. After understanding her concerns, Daniel introduces the "BUT U" app as a solution.

To make the app's benefits clear, Daniel simulates a road trip from Kuala Lumpur to Penang. He demonstrates how the app identifies charging stations along the route and filters for stations with rest areas and dining options. Daniel also highlights key features like real-time availability updates, customizable notifications, seamless payment integration, and an eco-points reward system. Reassured by the app's functionality, Sarah feels confident and decides to proceed with her EV purchase. Daniel reflects that the app not only helped secure the sale but also strengthened the dealership's reputation for excellent customer support.

Scenario 3: Planning a Road Trip (Siti Nuraini - The Road Tripper)

Excited for a weekend road trip from Johor Bahru to Penang, Siti can open the "BUT U" app to plan her journey. She can input her starting point and destination, and the app will suggest a route with reliable charging stations along the way. Siti can apply filters for stations with amenities like rest areas and restaurants, ensuring she has comfortable breaks during her drive.

The app also provides real-time station availability and allows her to reserve charging slots at the most convenient stops.

During the trip, Siti will receive notifications from the app about her selected charging stations, including updates on current availability and estimated wait times. At one stop, the app recommends a nearby café with Wi-Fi where she can relax while her car charges. Halfway through the trip, Siti may encounter unexpected traffic, and the app will dynamically adjust her route to include an alternative charging stop. This seamless experience will make Siti's road trip enjoyable and worry-free, reaffirming her love for traveling with an EV.